

High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 48 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on design simulation results obtained with typical external components specified in [Table 39](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 39. HSE oscillator characteristics

Symbol	Parameter ⁽¹⁾	Conditions ⁽²⁾	Min	Typ	Max	Unit
f_{OSC_IN}	Oscillator frequency	-	4	-	48	MHz
R_F	Feedback resistor	-	-	200	-	k Ω
$I_{DD(HSE)}$	HSE current consumption	During startup ⁽³⁾	-	-	13	mA
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 10\text{ pF@}8\text{ MHz}$	-	0.62	-	
		$V_{DD} = 3\text{ V}$, $R_m = 45\ \Omega$, $CL = 10\text{ pF@}8\text{ MHz}$	-	0.67	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 5\text{ pF@}48\text{ MHz}$	-	1.15	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 10\text{ pF@}48\text{ MHz}$	-	1.75	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 20\text{ pF@}48\text{ MHz}$	-	5.0	-	
G_m	Maximum critical crystal transconductance	Startup	-	-	1.5	mA/V
$t_{SU(HSE)}$ ⁽⁴⁾	Startup time	V_{DD} is stabilized	-	2	-	ms

1. Specified by design. Not tested in production.

2. Resonator characteristics given by the crystal/ceramic resonator manufacturer.

3. This consumption level occurs during the first 2/3 of the $t_{SU(HSE)}$ startup time

4. $t_{SU(HSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer

For C_{L1} and C_{L2} , it is recommended to use high-quality external ceramic capacitors in the 5 pF to 20 pF range (typ.), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator (see [Figure 19](#)). C_{L1} and C_{L2} are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C_{L1} and C_{L2} . PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing C_{L1} and C_{L2} .